



SECP2523: DATABASE (WBL)

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN

FOR *Koperasi KADA All-in-one Financial Platform*

Group Name: *Tech EcoPioneer*

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1.0 Introduction

Overview about the company

Koperasi KADA Online Platform System is a web platform responsive to mobile devices for the modern management of cooperatives, having an accessible financial system in place for the members. Members and administrators alike can log in to the system: members to manage their savings, loans, and transactions efficiently; admins for the control of all finances and data management. Program organizers may access detailed financial reports for a better decision on activities and an overview of such. The features of the platform will include automated validation of documents, secure OTP-based login, accessibility support, streamlined data storage, personalized dashboards, and efficient management to enhance user convenience and engagement. Thereby, the system will try to create an open collaborative and inclusive environment to instill trust and finance empowerment among cooperative members.

1.1 Problems

Lack of Integration

KADA Koperasi is not integrated with significant tools like accounting software, payment gateways, and banking platforms. This shortcoming has made administrators do some of the tasks manually, which include keeping financial records, processing loans, and reconciling payments. Due to the lack of automation, the risk of human error increases, such as data mismatches or omissions, but reduces in general efficiency. It further lacks integration, which creates a big problem in terms of scalability as the cooperative needs to accommodate a large number of members and transactions.

Evidence: The present system mostly relies on manual data transfer, which results in errors, delays, and inconsistencies that lead to inefficiencies in the management of cooperative operations.

Security Issues

The system has vulnerabilities to unauthorized access, data breaches, and financial fraud because it relies heavily on manual processes and lacks sophisticated security measures. Sensitive data,

not encrypted, with weak authentication mechanisms such as two-factor authentication (2FA), makes member information, which is personal and financial in nature, vulnerable. This lack of security exposes members to potential fraudsters but also taints the cooperative's reputation by reducing the confidence of members in its operations. In addition, poor access control means administrators cannot monitor or limit who accesses specific data, thereby increasing risks. Evidence: Cases of financial fraud and data breaches in other cooperatives using similar systems reveal, with crystal clarity, the vulnerability of manual frameworks such as that used by KADA. Members of other co-operatives report a sharp decline in trust when this sort of breach occurs.

Lack of Accessibility

The current system does not have any mobile access or online features for members and administrators to interface with the co-operative. Members are forced to come to the offices to make the simple deposits, check account balances, or even apply for loans. This prevents members from enjoying access to the digital ways due to their staying in the rural or remote areas with little or no computer or poor internet connectivity in the places where they stay. It effectively denies an overwhelming majority. It also lengthens the inconvenience suffered by members, since it is impossible to gain access to live financial updates.

Evidence includes the fact that those rural cooperatives that have adapted mobile-friendly platforms have increased participation and satisfaction by their members, while KADA's system remains dependent on physical contact, thereby leaving many members under-served and dissatisfied.

Static Member Engagement

There are no features in the current system that allow for dynamic engagement with the members, including real-time notifications, digital reminders, or interactive dashboards. It becomes hard for members to track the savings progress, loan status, or updates on the cooperative without physically visiting the office. This creates a distance between the cooperative and its members, which causes frustration and lessens the satisfaction level of the member. The lack of personalized notification means that members are not able to meet the critical deadline for loan repayments or any other financial commitments, which in turn complicates their experience further.

Evidence: Cooperative members normally suffer delays resulting from office visits to seek an update or transaction. Comparatively, cooperatives on systems with dynamic tools of engagement reported improved communication and retention of members.

1.2 Proposal

The Koperasi KADA Online Platform System is proposed to introduce automation and computerization in the management of a cooperative to handle the inefficiencies of the current

manual processes. This system introduces secure, accessible features that enhance the member experience and operational efficiency. It consists of five major modules: Authentication (Login), Member Registration, Manage Savings, Manage Loans, and Reporting-all tailored to streamline specific aspects of the operations of the cooperative.

The Login module covers the security and control in providing access to the system for members, administrators, and board of directors. This covers all basic features related to Login, Password Reset, and Account Registration with integration of advanced security features based on OTP via WhatsApp and encrypted credentials. The module implements role-based access control in distinguishing the different user roles where each user is granted access to only their relevant functionalities. In addition, an audit trail monitors the login activities regarding unauthorized access attempts, which enhances the security and reliability of the system.

The Member Registration module simplifies the onboarding process, whereby prospective members can apply to join online by submitting personal details and uploading required documents. Admins will then be able to review and approve these applications, thereby ensuring that new members meet the standards required by the cooperative. Members will then have access to personalized dashboards where they can manage their accounts and interact with the system. In this way, it avoids repetitive paper works, hence very efficient and transparent for both the member and administrators.

The **Manage Savings module** enables members to open savings accounts, set targets, and schedule repeat deposits. Members can track their savings performance in real time through their dashboards for increased financial transparency and accountability. Savings activities are overseen by admins to ensure that they are in line with the policies of the cooperative. It also supports automated interest accrual calculations to help members appreciate the growth of their savings while improving efficiency by reducing administrative work.

The **Manage Loans module** automates the whole application process, making it easier for members to access loans. Members will be able to apply, upload documents required for the loans, and estimate their repayment schedules using an integrated loan calculator. Admins are allowed to either approve or disapprove applications based on policies of the cooperative that guarantee fairness and speed. The module also keeps track of the repayment history, showing members and admins the status of loans transparently. This brings accountability and builds trust between members and the cooperative.

The Reporting Module provides tools for generating financial and operational reports. Admins and board directors can access data on member savings, loan disbursements, repayments, and overall system performance. These reports are customizable and can be exported for presentations or audits. The module provides charts and other graphic representations in visualizing data to enable the management of decision-making activities or pointing out valuable

insights into its operation. With this module, administrators will have comprehensive data to serve as a starting point for strategizing and wise decision-making activities.

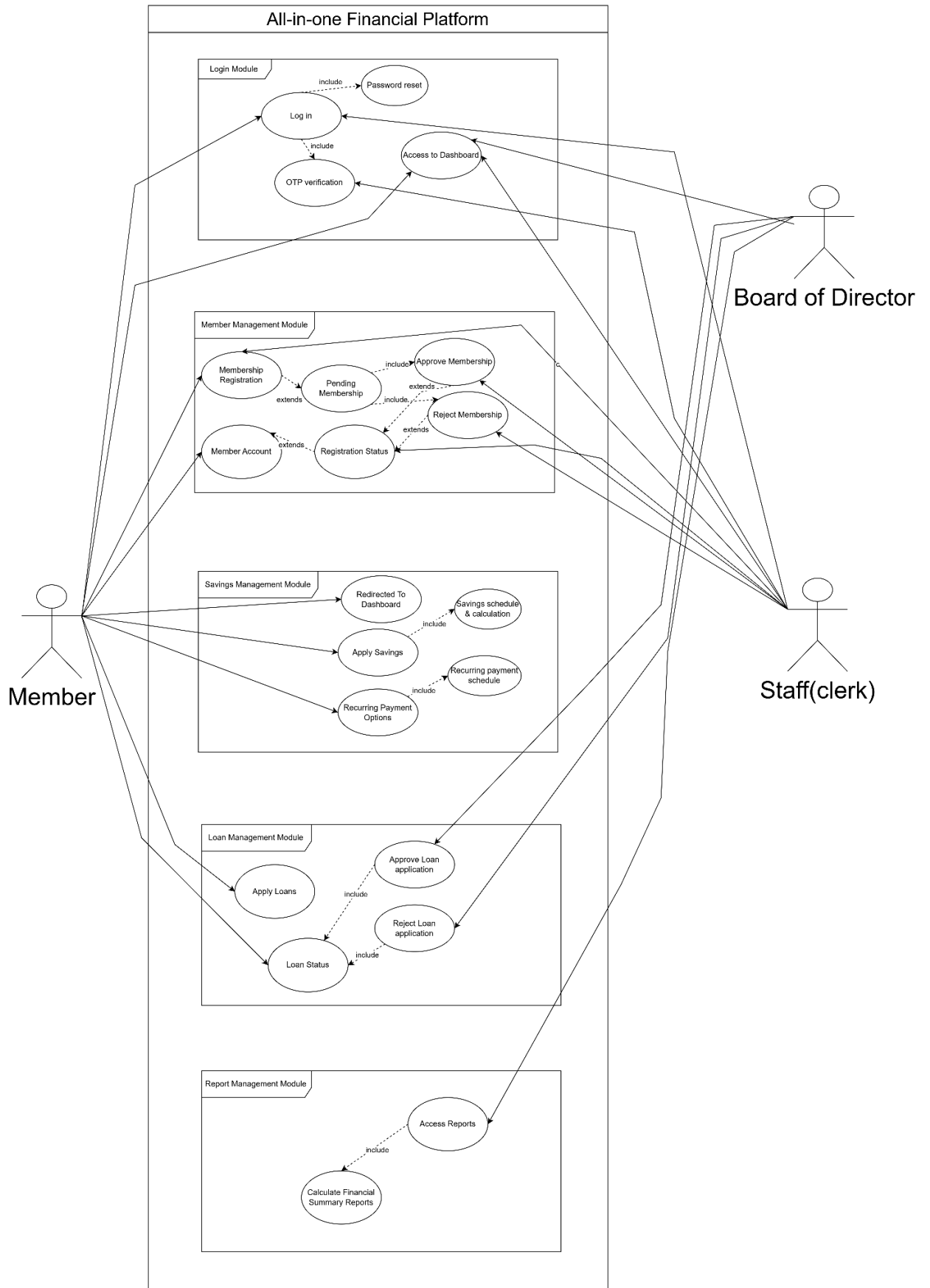
1.3 Definitions, Acronyms and Abbreviations

No	Acronyms/Abbreviations	Definition
1	Member	One of the targeted users in the system, they are members of the cooperative using the system to manage their savings, loans, and account details.
2	Admin	One of the targeted users in the system, they are administrators responsible for maintaining the system, approving member registrations, and overseeing financial operations like savings and loans.
3	Savings Account	One of the entities in the system. It represents the members' savings, allowing deposits, recurring payments, and tracking of accumulated funds.
4	Loan Application	One of the entities in the system. It is specific for members applying for financial assistance, involving submission, review, and approval processes.
5	Reporting	One of the features in the system that generates detailed summaries of financial activities, including savings, loans, and overall cooperative performance for admin use.
6	OTP (One-Time-Password)	A security feature used during login to authenticate users via a unique password sent to their mobile device or email.S

2.0 Database System Requirement

2.1 System Overview

The Koperasi KADA Online Platform System is an integrated cloud-based financial management solution that could be used for streamlining or digitizing cooperative operations. This platform will enable members to register, manage savings accounts, apply for loans, and set recurring savings payments through a user-friendly interface. Staff will be able to validate documents, approve applications for registration, process loans, and manage customer support tickets. The directors shall have the capability to access financial reports, approve loans, and monitor system performance. Other significant features include OTP-based login through WhatsApp, automated notifications via email, transaction history, and a user assistance chatbot. It will not support non-financial cooperative operations like inventory or retail management. Its main concentration will be financial and member-related functions with a view to optimizing workflow and enhancing user experience within cooperatives. This platform is especially fitted for the application to iron out some inefficiencies from the old manual cooperative system by automating critical processes that reduce administrative workload and improve security. Among its objectives are financial inclusivity, transparency, and user satisfaction through digital tools in an accessible and efficient manner. The major benefits derived from it are smooth operation efficiency, reduced errors, increasing member engagement, and scalability for the growth of the cooperative. This specification meets higher-level requirements and system objectives, providing a secure, scalable, and user-friendly solution for cooperative financial management.



2.2 Module 1: Login

Dheshieghan A/L Saravana Moorthy is the person in charge of documenting this module.

2.2.1 Module Description

Login is a core module of the system that provides secure access to Members and Administrators. This module can be used for user login and password reset. The Login feature will let any user securely log into their personalized dashboard upon verification of his/her credentials like email and password against stored data. If one forgets his/her password, the system has a secure password reset mechanism. In such cases, users can initiate a password reset process by entering their registered email address, getting a reset link or OTP, and setting a new password securely. Account registration includes the process of registering new members into the system, where personal information, password creation, and verification via email or OTP take place. The super-administrators create and manage accounts for administrators; this is a role-based access control for added security.

This module is designed to keep security at its best and user interaction seamless. It includes different validation techniques, such as verification of email and strength of password, to minimize errors and enhance the integrity of the user data. It also further allows the checking of failed login attempts, ensuring countermeasures such as account lockout or using CAPTCHA for preventing unauthorized access. An authentication module which logs login activity by Members and Administrators, visible by super-administrators for audit and debugging reasons. In summary, this module provides not only a secure entrance to the system but also an improved user experience by making the authentication process smooth and easy for the users.

2.2.2 Transaction Requirement: Data Entry

The Login module provides a means for logging in, and password recovery in an efficient and secure manner. Members log in with their email and password, while new accounts require additional information such as name and email that need to be verified for correctness and according to security standards. Password recovery allows the user to input their registered email address in order to receive a reset link, at the end of which they can set a new secure password. These ensure accurate, secure, and user-friendly data handling.

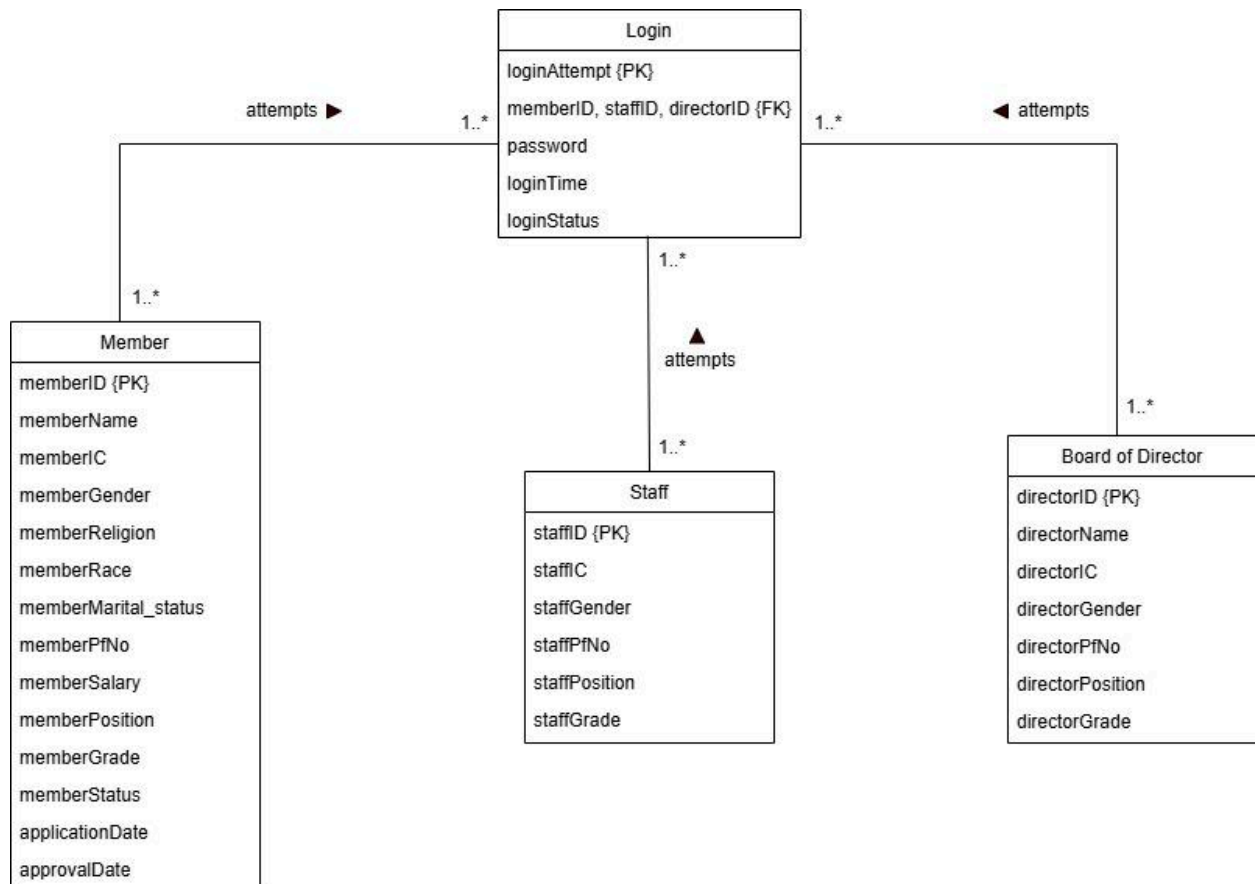
2.2.3 Transaction Requirement: Data Update/Deletion

The Login module grants users a secure interface to change or delete their credentials, such as email and password. Members can edit the account details with the rationale of keeping information up to date and may opt to delete the accounts, thus having control over their personal information within the system.

2.2.4 Transaction Requirement: Data Queries

1. List of users with their username/email, registration date, account status, and role.
2. List of members with their personal details, registration status, and uploaded documents.
3. List of failed login attempts for admin monitoring purposes.

2.2.5 Local Conceptual Data Model (ERD)



2.3 Module 2: Member Registration

Evelyn Goh Yuan Qi is the person in charge of documenting this module.

2.3.1 Module Description

The Member Registration module is a critical component of the cooperative system, offering seamless interactions between users and staff. It streamlines the registration process for new users, ensuring a user-friendly and efficient experience while maintaining the accuracy and security of member data. One of the use cases is Membership Registration. Through an online membership registration form, users can provide their personal details, such as name, phone number, address, and so on. The system validates the data for completeness and correctness before submission. After completely submitting the application, the system stores the application with a status of 'Pending'.

The next use cases are Approve and Reject Membership. Staff members play a key role in this module, using a dedicated interface to review and validate the submitted applications. Based on their evaluation, staff can approve or reject the applications. Notifications of the application status are then sent to users. Once the applications are approved by the staff, members can open and manage accounts for loan and saving. This module provides an effective foundation for onboarding new users and advancing the cooperative's mission by promoting inclusivity, scalability, and secure data management.

2.3.2 Transaction Requirement: Data Entry

The purpose of this transaction is to collect and store member application details. Users can complete the membership registration application by entering their personal details and necessary information. After completing the application, users can submit it. Next, after submitting the application, the data will be stored in the database and the application status is set to 'Pending'.

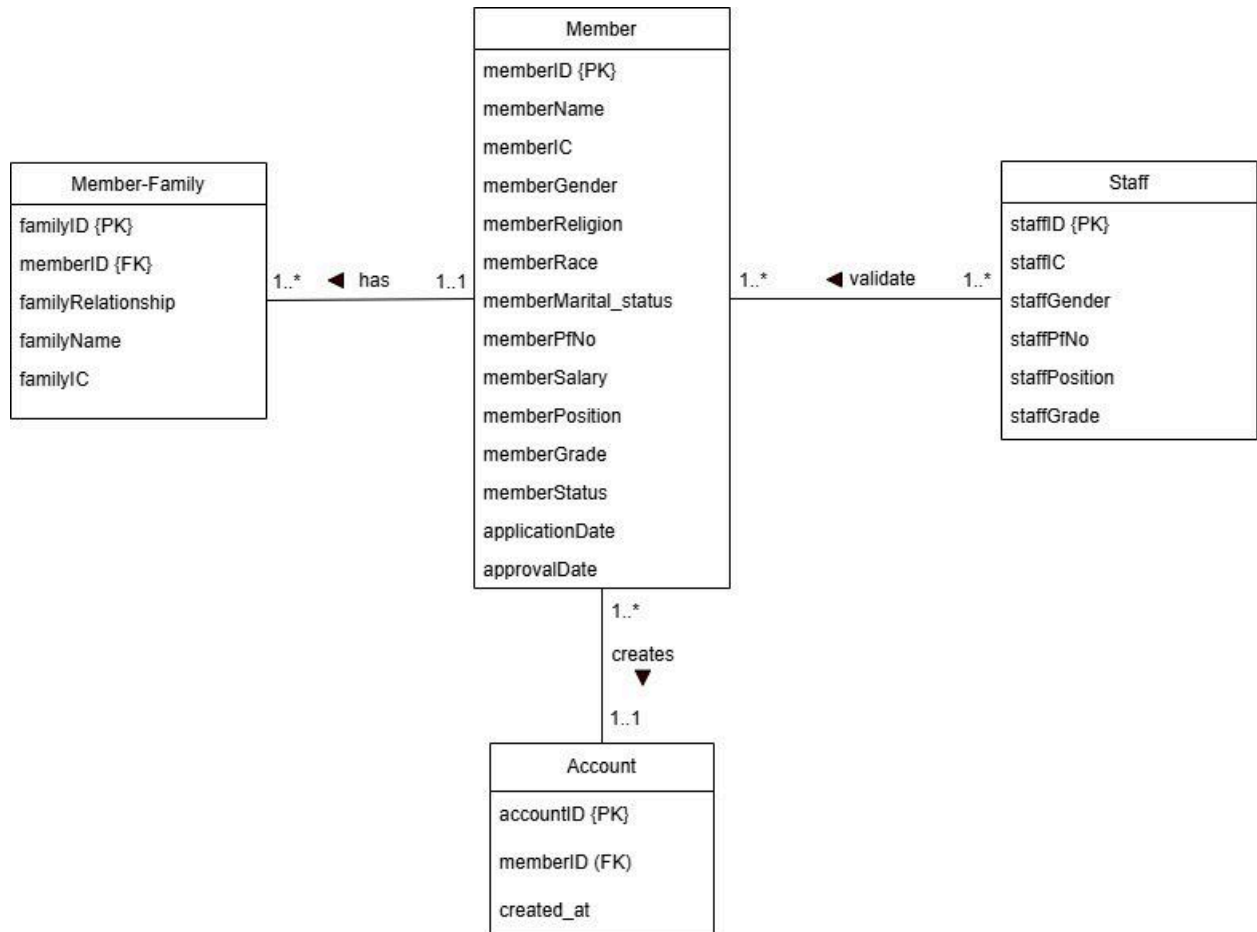
2.3.3 Transaction Requirement: Data Update/Deletion

No data updates are included in this module. For data deletion, staff can delete duplicate or incomplete applications and the rejected applications are archived for record-keeping.

2.3.4 Transaction Requirement: Data Queries

1. The list of pending applications can be retrieved by staff.
2. The list of approved applications can be retrieved by staff.

2.3.5 Local Conceptual Data Model (ERD)



2.4 Module 3 <Saving>

CHEW CHIU XIAN is the person in charge of documenting this module.

2.4.1. Module Description

This module will involve actor member of KADA only. There are 3 use cases inside this module: UC001: Create Saving Goals and UC002: Recurring Payment Option and UC003: Save Transaction History.

UC001: Create Saving Goals

- In this module, member of KADA able to set saving goals.
- Member can create and edit saving goal directly in the website.
- Member shall provide the details such as target saving amount, duration such as in month and initial deposit, which is optional while applying saving account.
- The system will save the savings goal details in the database and the dashboard will be updated to show new saving goals.

UC002: Recurring Payment Option

- In this use case, members of KADA are able to automate savings deposits by setting up recurring payment schedules.
- Members shall enter details such as payment frequency like monthly or weekly, amount of payment and preferred payment method, either bank transfer or credit card payment in order to set up recurring payment schedules.
- The system will display confirmation message to the member to confirm the recurring payment details.
- After confirmation, user's dashboard is updated with the recurring payment setup details.

UC003: Save Transaction History

- In this use case, every transaction and payment done by member of KADA will be saved by the system.
- The system will show list of transaction done by member of KADA in the dashboard.

2.4.2. Transaction Requirement: Data Entry

UC001: Create Saving Goals

Members provide the following details:

- Target savings amount
- Duration (in months)
- Initial deposit (optional)

UC002: Recurring Payment Option

Members input:

- Payment frequency (e.g., monthly, weekly)
- Payment amount
- Preferred payment method (e.g., bank transfer, credit card)

UC003: Save Transaction History

Every transaction process provide:

- Amount of money to be transacted
- Type of transaction
- Method for Transaction

2.4.3 Transaction Requirement: Data Update/Deletion

UC001: Create Saving Goals

- Members can update savings goals (target amount or duration)
- Members can delete savings goals (with restrictions based on balance and existing commitments)

UC002: Recurring Payment Option

- Members can update recurring payment details (frequency, amount, method)
- Members can delete recurring payment schedules

UC003: Save Transaction History

- The system updates the transaction history list for every transaction done

2.4.4 Transaction Requirement: Data Queries

UC001: Create Saving Goals

Members can query saving goal progress.

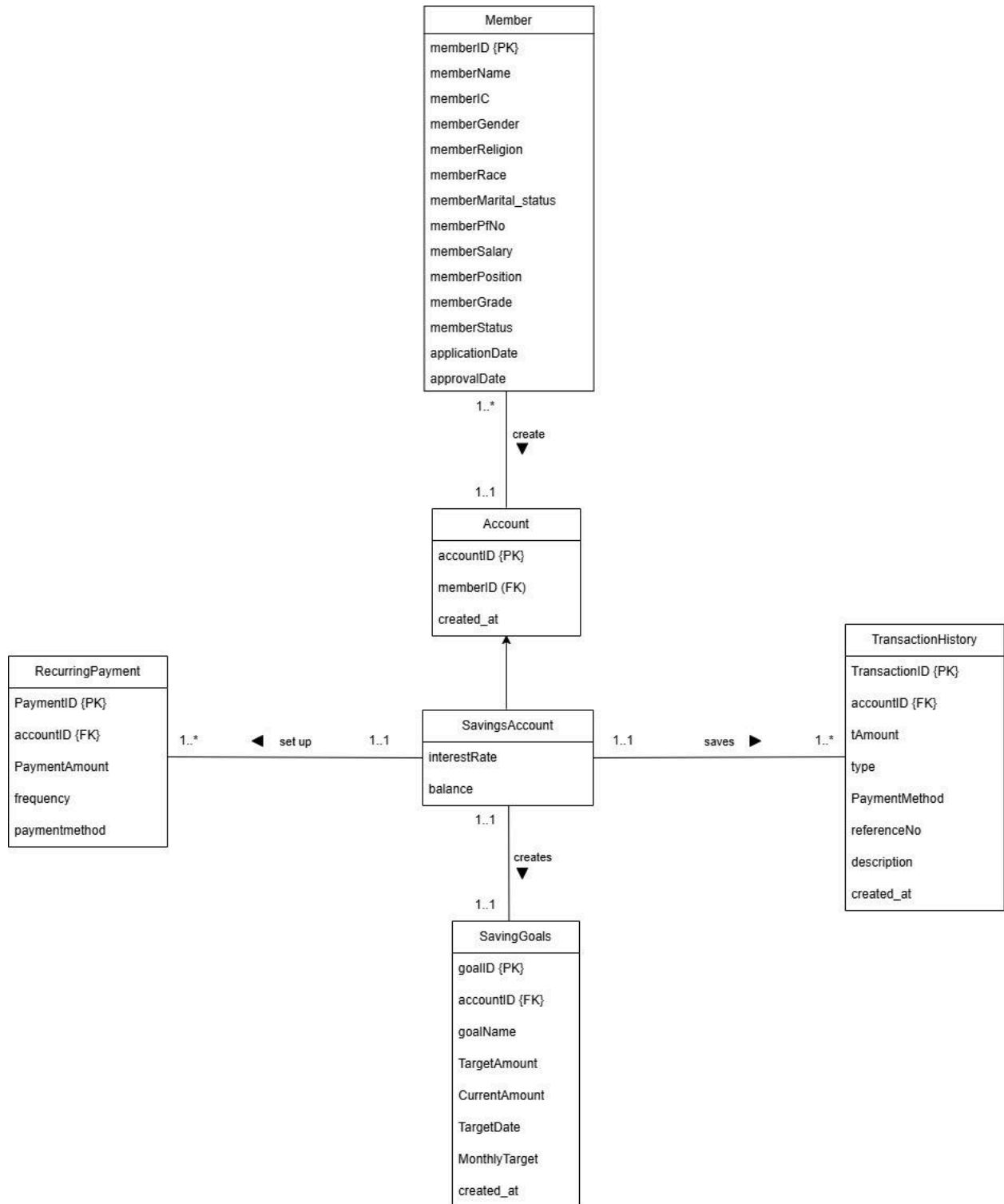
UC002: Recurring Payment Option

- a) Members can query scheduled recurring payments
- b) Staff can query reports on recurring payment trends

UC003: Save Transaction History

The system query the transaction history list in the dashboard

2.4.5 Local Conceptual Data Model (ERD)



2.5 MODULE 4 - Loan

Ng Yu Hin is the person in charge of documenting this module.

Module description

This module involves the member and board of directors. Members can request a loan by submitting a loan application while board of directors are responsible for approval or rejection of said request. Members can use this loan application to suggest the amount requested and period of time suggested for monthly installment. After completely submitting the loan application, the system stores the application with a status of 'Pending'. The board of directors will review and decide whether member is valid for loans. If rejected, members can retry the request again until a successful attempt. Once approved, the amount of loan will be recorded by the system which would be used for generating report purposes and members are required to pay the monthly installment. The main feature of this module is to request loans by members and admins approving or rejecting as well as a build in calculator for calculating the monthly installment according to a same constant rate which is 4.2%.

Transaction Requirement: Data Entry

1. Input the required information such as personal details
2. Add new data records of loan applications details to the database

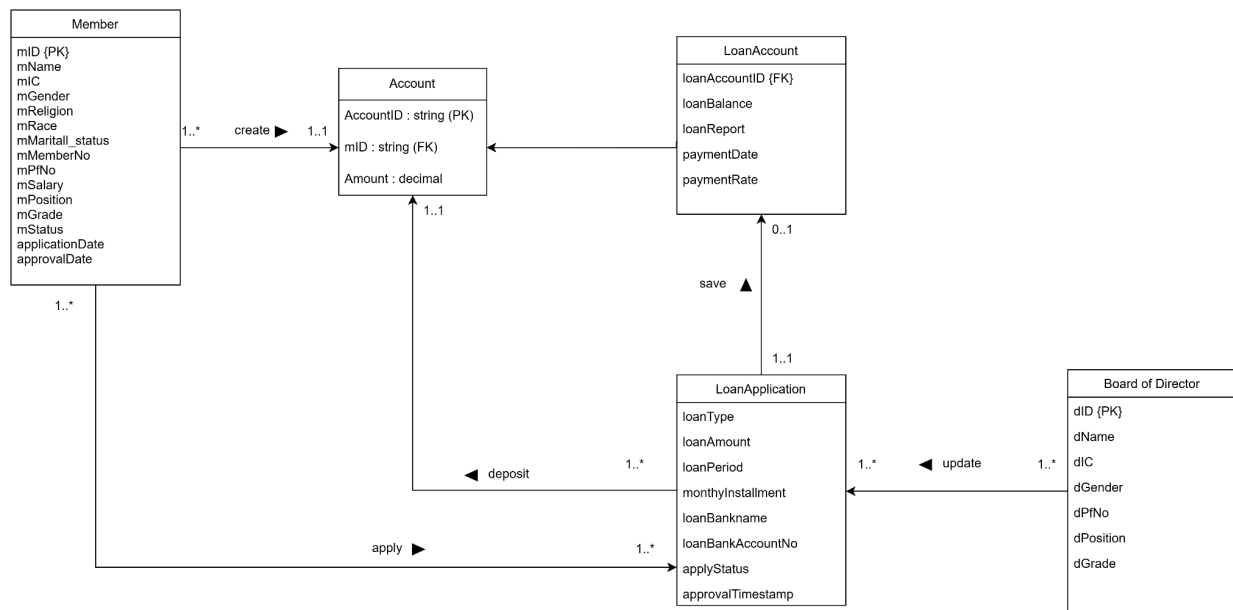
Transaction Requirement: Data Update/Delete

1. Update status of requested loan application
2. Update account balance from applying loans
3. Delete rejected or incomplete loan applications

Transaction Requirement: Data Queries

1. Board of director can query and view all forms submitted by members for pending approval or rejections
2. Generate simple report for member and board of directors

Local Conceptual Data Model (ERD) for Module 4



2.6 MODULE 5 - Reporting

Lau Yan Kai is the person in charge of documenting this module.

Module description

This reporting module involved staff and board of directors. It is used to help users generate, view and interact with the reports. The report provides a clear and detailed picture of the organisation's performance. Users can use these reports to understand the financial health of the organisation, track operational progress, and make decisions based on the real data. Key features of this module include the ability to view reports, create custom reports based on the specific filters, and export or print reports for future reference. This ensures that the stakeholders have access to accurate information.

This module helps users with several functions. The users can open reports that show things like financial performance, trend, or operational activities. If specific information is needed, the user can create a custom report by selecting filters such as date ranges or types of report. Once the report is ready, it can be saved as a file, either in PDF, Excel or CSV, or be printed for sharing. The module also checks that the data used in reports is correct and complete. If something is missing or wrong, it will alert the user so they can fix it.

Transaction Requirement: Data Entry

1. Input filters or parameters for custom reports, like the start and end dates, or specific metrics
2. Add new data records or manual adjustments to the database

Transaction Requirement: Data Update/Delete

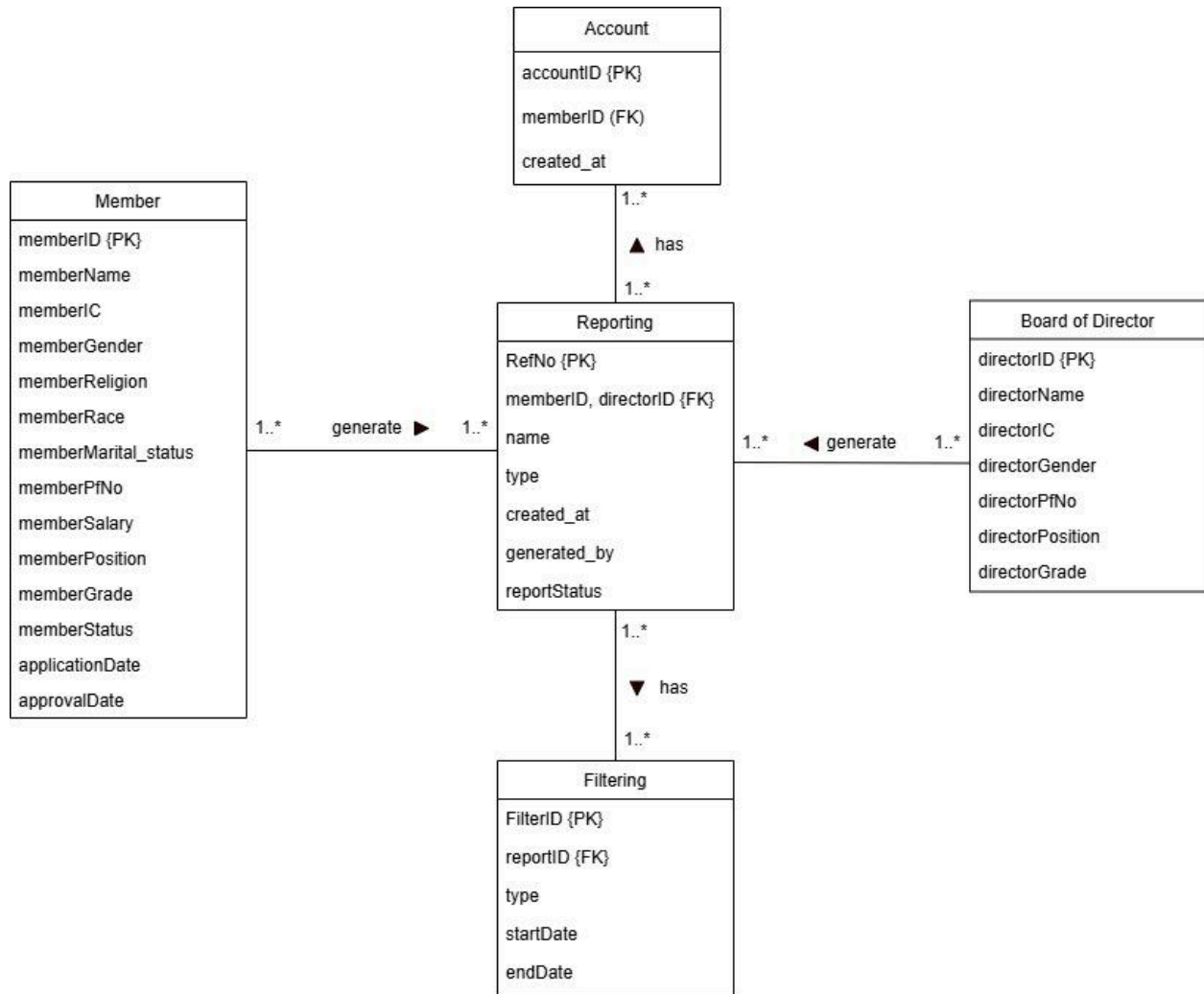
1. Update the report details, such as changing the name, or any parameters
2. Modify data record
3. Delete old or incorrect data records
4. Remove outdated or unused reports from the database

Transaction Requirement: Data Queries

1. Retrieve financial summaries for a selected period
2. Generate reports for a specific time period or range
3. Fetch records that match specific filters or parameters
4. Retrieve data from multiple sources

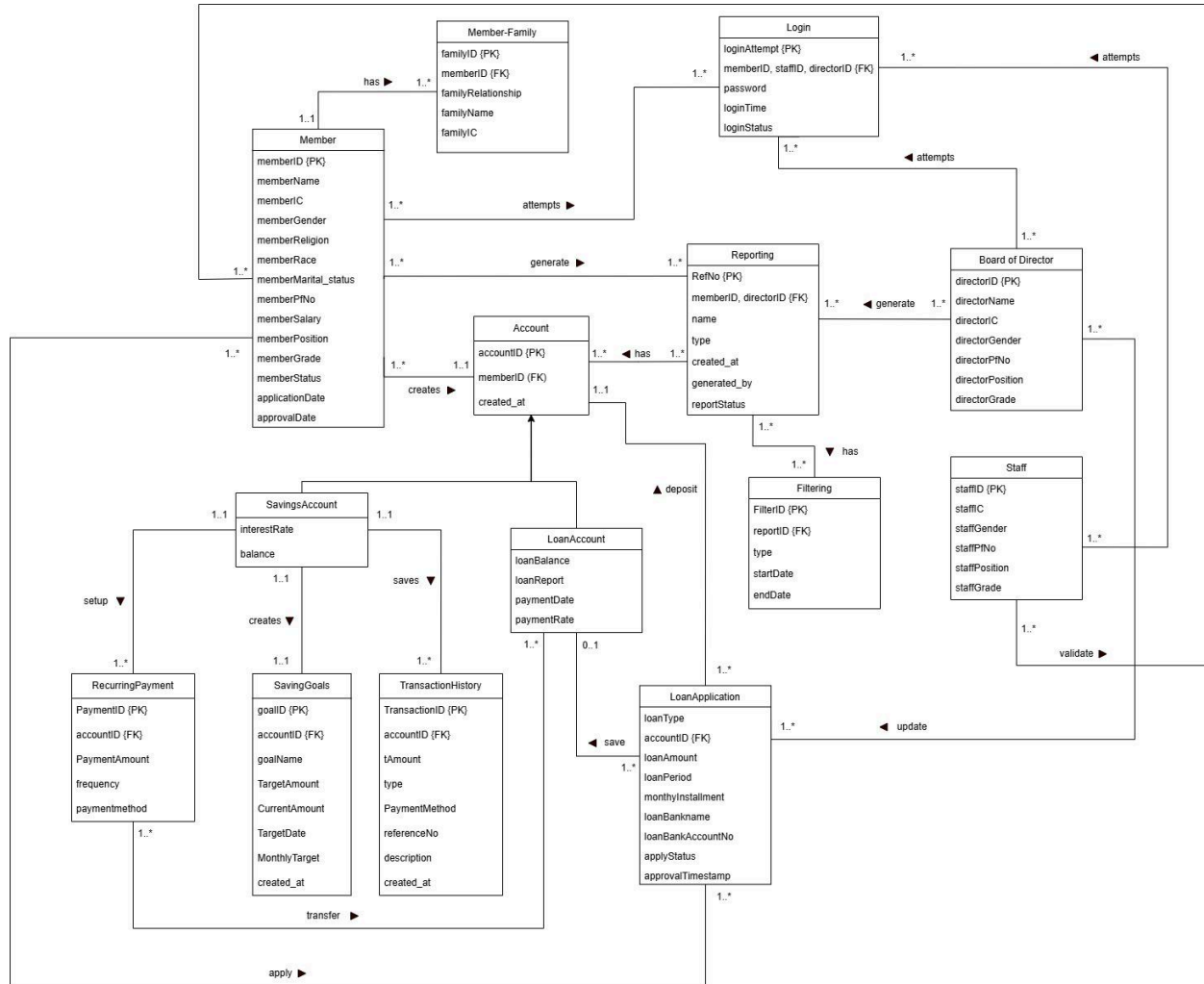
- Run custom queries based on user-defined criteria such as comparing revenue trends

Local Conceptual Data Model (ERD)



Global Conceptual Data Model (Global ERD)

3.1 Global Conceptual Design (ERD)



3.2 Verified Global Conceptual Design (Verified ERD)

3.3 Data Dictionary for the global conceptual ERD

3.3.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
Member			
Staff			
Board of director			
Login			
Member-Family			
Account			
SavingsAccount			
RecurringPayment			
SavingGoals			
TransactionHistory			
LoanAccount			
LoanApplication			
Reporting			
Filtering			

3.3.2 Identifying Relationship Types

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Member	1..*	attempts	Login	1..*
Board of Director	1..*	attempts	Login	1..*
Staff	1..*	attempts	Login	1..*
Staff	1..*	validate	Member	1..*
Member	1..1	has	Member-family	1..*
Member	1..*	creates	Account	1..1
SavingsAccount	1..1	setup	RecurringPayment	1..*
	1..1	creates	SavingGoals	1..1
	1..1	saves	TransactionHistory	1..*
RecurringPayment	1..*	transfer	LoanAccount	1..*
Member	1..*	apply	LoanApplication	1..*
LoanApplication	1..*	save	LoanAccount	0..1
	1..*	deposit	Account	1..1
Board of Director	1..*	update	LoanApplication	1..*
Member	1..*	generate	Reporting	1..*
Board of director	1..*	generate	Reporting	1..*
Reporting	1..*	has	Account	1..*
	1..*	has	Filtering	1..*

3.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description
Member	memberID (PK)	int	Member's ID
	memberName	varchar (255)	Member's name
	memberIC	varchar(22)	Member's IC
	memberGender	enum('Male','Female')	Member's gender
	memberReligion	varchar(50)	Member's Religion
	memberRace	varchar(50)	Member's Race
	memberMarital_status	enum('Single','Married','Divorced','Widowed')	Member's marital status
	memberPFNo	int	Member's PF no
	memberSalary	decimal(10,2)	Member's salary
	memberPosition	varchar(100)	Member's position
	memberGrade	varchar(50)	Member's grade
	memberStatus	enum('Pending','Lulus','Tolak')	Member's membership registration status
	applicationDate	timestamp	Application date of the membership registration
	approvalDate	timestamp	Approval date of the membership registration
Staff	staffID (PK)	int	Staff's ID
	staffC	varchar(22)	Staff's IC
	staffGender	enum('Male','Female')	Staff's gender
	staffPfNo	int	Staff's PF no
	staffPosition	varchar(100)	Staff's position

	staffGrade	varchar(50)	Staff's grade
Board of Director	directorID (PK)	int	Board of Director's ID
	directorName	varchar (255)	Board of Director's Name
	directorIC	varchar(22)	Board of Director's IC
	directorGender	enum('Male','Female')	Board of Director's Gender
	directorPfNo	int	Board of Director's PF No
	directorPosition	varchar(100)	Board of Director's position
	directorGrade	varchar(50)	Board of Director's grade
Member-Family	familyID (PK)	int	Family member's ID
	memberID (FK)	int	Member's ID (refer Member entity)
	familyRelationship	varchar (50)	Family member's relationship
	familyName	varchar (255)	Family member's name
	familyIC	varchar (22)	Family member's ic
Account	accountID (PK)	int	Account's ID
	memberID (FK)	int	Member's ID(refer Member entity)
	created_at	timestamp	Timestamp of when the account was created
Login	loginAttempt (PK)	int	Unique identifier for each login attempt.
	memberID,	int	Member's ID (refer

	staffID, directorID (FK)		Member entity), staff's ID (refer Staff entity), director's ID(refer Board of Director entity)
	password	varchar(100)	Account password for login
	loginTime	timestamp	Time for login
	loginStatus	enum('Successful','Fail')	Status of login
LoanAccount	loanBalance	decimal(10,2)	Balance of the loan
	loanReport	varchar (255)	Report details of the loan (if applicable)
	paymentDate	date	Date of the next payment due
	paymentRate	decimal(5,2)	Rate of payment for the loan
LoanApplication	loanType	varchar (100)	Type of loan
	accountID (FK)	int	Account's ID (refer Account entity)
	loanAmount	decimal(10,2)	Total amount of the loan applied for
	loanPeriod	int	Loan repayment period in months
	monthlyInstallment	decimal(10,2)	Monthly installment amount for the loan
	loanBankname	varchar (100)	Bank account name associated with the loan
	loanBankAccount No	int	Bank account number associated with the loan
	applyStatus	enum('Pending','Lulus',' Tolak')	Status of the loan application

	approvalTimestamp	timestamp	Timestamp of when the loan application was approved or rejected
SavingsAccount	interestRate	decimal(10,2)	Saving account's interest rate
	balance	decimal(10,2)	Current balance in the saving account
RecurringPayment	PaymentID (PK)	int	Recurring payment's ID
	accountID (FK)	int	Account's ID (refer Account entity)
	PaymentAmount	decimal(10,2)	Amount of the payment
	frequency	varchar (100)	Frequency of the recurring payment
	paymentmethod	varchar (100)	Method used for payment
SavingGoals	goalID (PK)	varchar (100)	ID of the saving goal
	accountID (FK)	int	Account's ID (refer Account entity)
	goalName	varchar (100)	Name of the saving goal
	TargetAmount	decimal(10,2)	Target amount of saving
	CurrentAmount	decimal(10,2)	Current amount of saving
	TargetDate	date	Target date to achieve the saving goal
	MonthlyTarget	decimal(10,2)	Monthly target of the saving goal
	created_at	timestamp	Timestamp of when the saving goal was

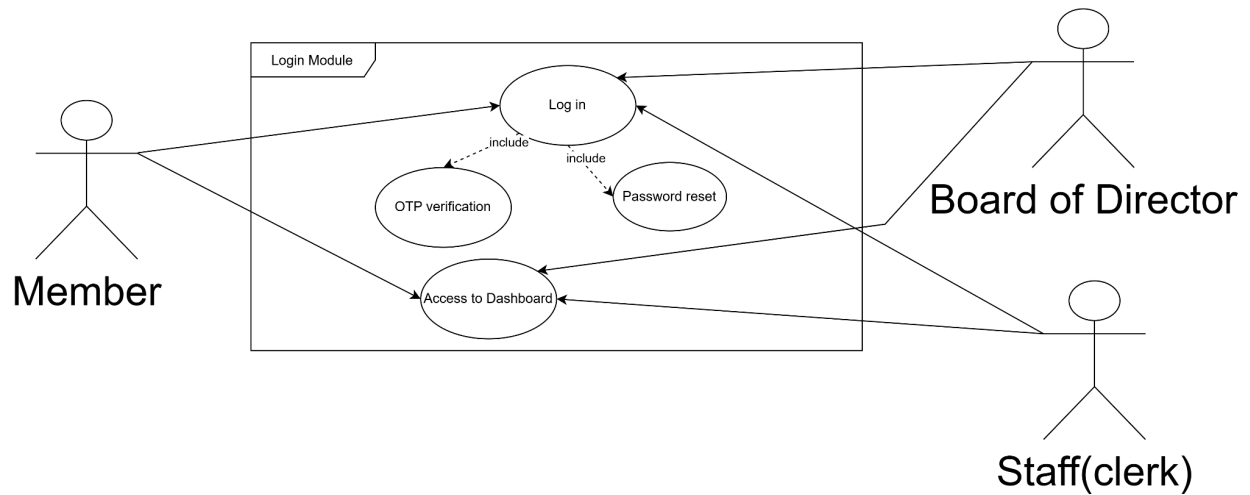
			created
TransactionHistory	TransactionID (PK)	int	Transaction history's ID
	AccountID (FK)	int	Account's ID (refer Account entity)
	tAmount	decimal(10,2)	Transaction's amount
	type	enum("deposit", "withdrawal", "transfer_in", "transfer_out")	Type of transaction
	PaymentMethod	enum("bank_transfer", "salary_deduction", "fpx", "credit_card")	Payment method of transaction
	referenceNo	varchar (100)	Reference no of transaction
	description	text	Description of transaction
	created_at	timestamp	Timestamp of when the transaction was made
Reporting	RefNo (PK)	int	Reference no of report
	memberID, directorID (FK)	int	Member's ID (refer member entity), director's ID(refer Board of Director entity)
	name	varchar (100)	Name of report
	type	varchar (100)	Type of report
	created_at	timestamp	Timestamp of when the report was created
	generated_by	Varchar (100)	Name or identifier of the person that generated the report

	reportStatus	enum	Status of the report
Filtering	FilterID (PK)	int	Unique identifier for the filter
	reportID (FK)	int	Report's ID (refer Reporting entity)
	type	varchar (100)	Type of filtering applied
	startDate	Date	Start date for the filter range
	endDate	Date	End date for the filter range

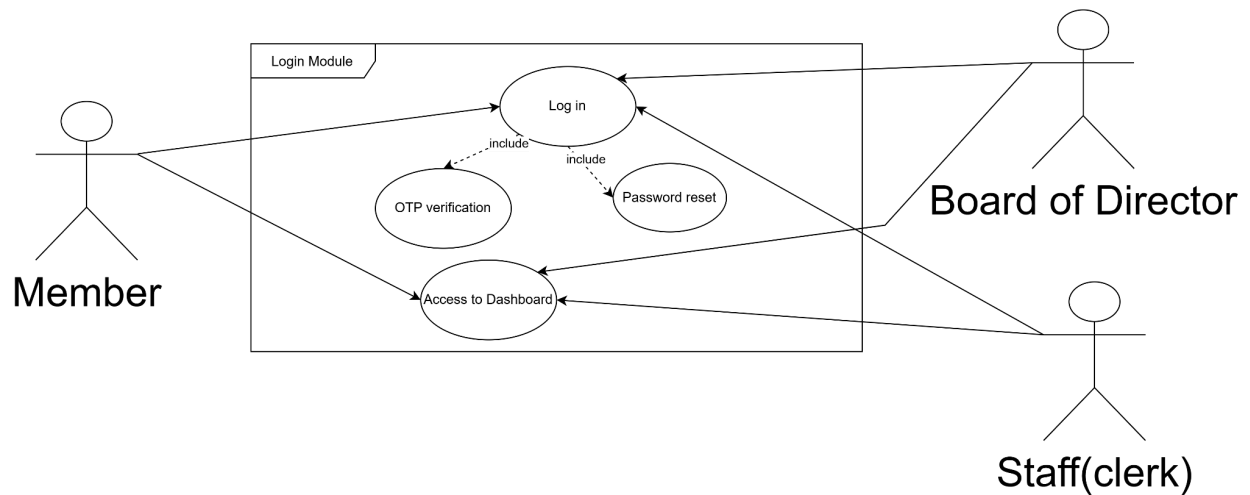
Appendices

4.1 Use Case Diagram for system

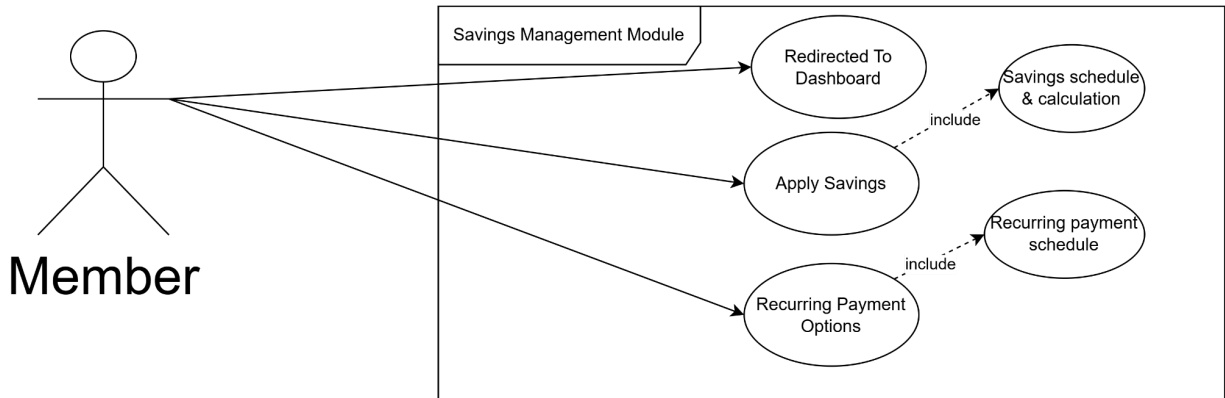
Authentication Use Case Diagram



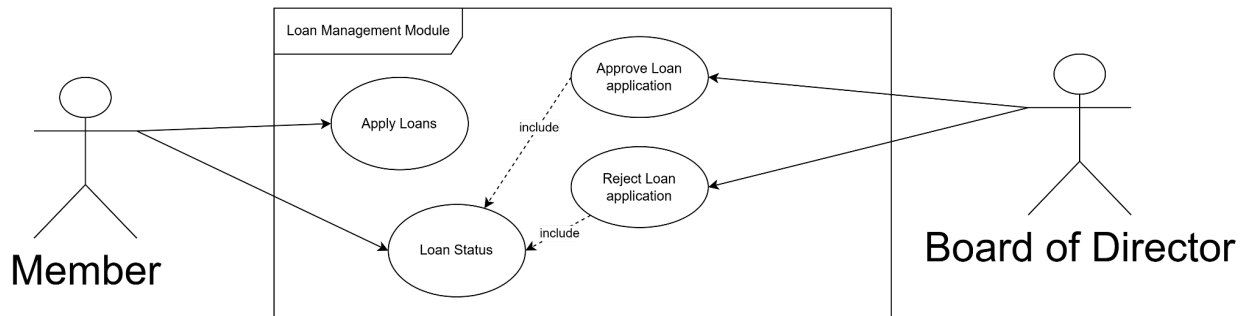
Member Registration Use Case Diagram



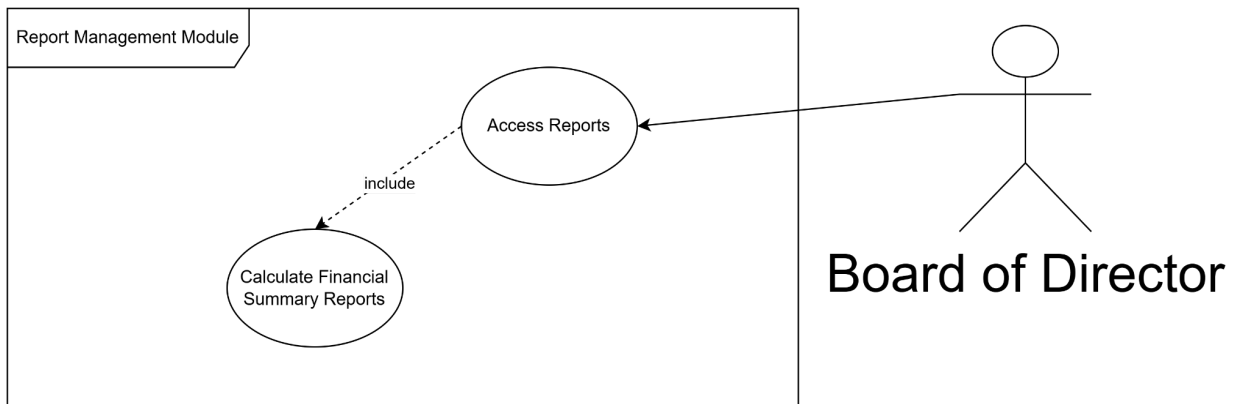
Savings Use Case Diagram



Loan Use Case Diagram



Report Use Case Diagram



4.2 Activity Diagrams

4.3 Meeting Log